

MERIDEN MARKHAM, CT, USA

WMO: 725027

Lat: 41.510N

Lon: 72.828W

Elev: 31

StdP: 100.95

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54788

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-12.4	-23.0	0.5	-12.7	-20.7	0.6	-9.5	10.3	-2.6	9.0	-3.3	1.7	320	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.5	23.1	31.1	22.4	29.5	21.5	24.7	30.0	23.9	28.5	23.1	27.1	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	17.9	26.6	22.5	17.3	26.0	22.0	16.7	25.5	75.1	30.1	71.7	28.5	68.7	27.1	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.5	7.5	6.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-19.5	35.3	3.1	1.9	-21.7	36.6	-23.5	37.7	-25.2	38.8	-27.5	40.1		
			-19.9	26.2	3.0	0.8	-22.0	26.8	-23.8	27.3	-25.4	27.8	-27.6	28.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-2.2	-1.1	3.4	9.7	15.4	19.9	23.4	22.4	18.6	12.0	6.0	1.0
	DBStd	9.87	5.68	5.05	4.93	4.60	4.03	3.43	2.68	2.84	3.58	4.58	4.49	4.88
	HDD10.0	1414	380	312	212	58	3	0	0	0	0	32	134	283
	HDD18.3	3193	638	544	464	260	110	22	1	2	41	203	369	538
	CDD10.0	1697	1	1	7	51	171	298	415	384	258	93	16	3
	CDD18.3	434	0	0	0	3	20	71	158	128	49	6	0	0
	CDH23.3	3808	0	0	1	49	227	628	1418	1080	371	34	1	0
	CDH26.7	1231	0	0	0	16	63	193	519	345	90	4	0	0
Wind	WSAvg	2.4	2.7	2.8	3.2	3.1	2.5	2.2	1.9	1.8	2.0	2.2	2.4	2.5
Precipitation	PrecAvg	1336	98	81	119	106	101	109	113	117	129	126	115	114
	PrecMax	1655	188	247	220	231	210	253	223	230	351	414	189	204
	PrecMin	1010	48	41	18	23	13	18	45	38	41	15	36	35
	PrecStd	161	43	40	50	51	48	65	45	61	80	94	35	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.9	20.9	29.1	31.2	32.9	34.7	33.9	32.2	26.9	20.6	16.8
		MCWB	11.3	11.5	13.0	18.3	20.6	23.8	24.1	24.5	23.1	19.9	15.2	13.9
	2%	DB	11.3	12.0	17.0	23.7	28.4	30.9	32.8	31.9	29.0	23.8	17.9	13.5
		MCWB	8.8	8.7	10.5	14.2	19.3	22.2	23.3	22.9	20.9	18.5	14.1	11.6
	5%	DB	7.9	8.8	13.6	20.6	25.9	28.8	31.3	30.1	27.2	21.9	16.2	11.3
		MCWB	5.4	5.3	8.5	12.6	17.6	20.6	22.7	22.2	20.1	17.8	12.9	9.1
	10%	DB	5.8	6.2	11.1	17.8	23.2	27.2	29.6	28.5	25.4	19.7	14.0	8.8
		MCWB	3.2	3.1	7.1	11.2	16.3	19.9	21.9	21.4	19.3	16.1	10.9	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	12.6	14.8	18.7	22.2	24.7	26.0	26.1	24.2	21.9	17.4	14.4
		MCDB	13.4	14.0	18.4	28.6	29.3	31.1	32.2	32.2	29.1	24.6	19.0	15.6
	2%	WB	8.9	9.0	11.9	16.5	20.6	23.6	24.9	24.7	23.0	20.2	15.4	11.9
		MCDB	10.4	11.4	14.6	21.2	25.8	28.9	30.5	29.2	26.4	22.5	17.1	13.1
	5%	WB	6.0	5.7	9.5	14.5	19.3	22.4	24.0	23.8	22.0	18.5	13.7	9.5
		MCDB	7.8	8.1	13.0	18.0	24.0	26.8	28.6	27.8	25.1	21.0	15.6	11.4
	10%	WB	3.4	3.4	7.6	12.4	17.8	21.3	23.2	22.9	21.0	16.8	11.5	6.5
		MCDB	5.3	5.8	10.5	16.1	21.6	24.8	27.3	26.3	23.8	19.2	13.5	8.2
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	10.2	11.5	11.7	11.1	11.0	11.0	11.5	11.1	10.5	8.9
		MCDBR	11.6	13.2	14.5	15.7	15.7	13.8	13.0	12.4	13.2	12.9	13.0	11.3
		MCWBR	9.6	9.7	9.0	7.8	7.5	6.2	5.0	5.1	6.3	7.6	9.4	9.8
	5% WB	MCDBR	11.2	12.1	12.4	13.0	12.9	11.8	10.6	10.2	10.0	10.4	11.0	10.8
		MCWBR	10.0	10.0	9.1	7.5	7.0	6.0	4.8	4.9	5.9	7.2	9.5	10.2
Clear-Sky Solar Irradiance	taub	0.305	0.316	0.340	0.373	0.420	0.465	0.484	0.455	0.383	0.346	0.317	0.303	
	taud	2.455	2.433	2.430	2.368	2.263	2.175	2.133	2.206	2.380	2.498	2.537	2.531	
	Ebn at Noon	849	894	910	897	858	818	797	809	850	849	832	824	
	Edh at Noon	77	91	103	117	134	146	151	137	107	85	71	66	
All-Sky Solar Radiation	RadAvg	1.86	2.72	3.70	4.78	5.34	5.81	5.95	5.27	4.35	2.91	2.04	1.54	
	RadStd	0.15	0.24	0.28	0.52	0.53	0.57	0.33	0.30	0.36	0.28	0.16	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page